4A 1 WHAT IS ORGANIC CHEMISTRY?

SPECIFICATION REFERENCE 4.9

LEARNING OBJECTIVES

- Understand that alkanes and cycloalkanes are hydrocarbons which are saturated.



▲ **fig A** Friedrich Wöhler was a German chemistry professor and an early pioneer in the field of organic chemistry. He is best remembered for making urea (an organic compound) starting only from inorganic compounds.

EARLY DAYS

Since the 1800s, our knowledge and understanding of chemistry has grown rapidly. To make sense of all this knowledge, chemists divided chemistry into three main categories: inorganic, organic and physical chemistry. Each category is equally important. There are millions of different compounds in existence, and the vast majority are organic compounds. Now, what is organic chemistry?

Today, the word 'organic' has a very different meaning in everyday life, and it is often applied to farming and food. Going back to the 1800s, people believed that there was something special about some substances. They were only made in plants or animals. One example is the compound called urea, which is present in human urine. People used to believe that it could only be produced in the human body. In 1828, the German chemist Friedrich Wöhler discovered that urea could be made by heating a compound (ammonium cyanate) that was not organic. This meant that the idea of organic compounds only coming from living things was no longer correct.

HYDROCARBONS

The main feature of an organic compound is that it contains carbon. Almost all of these compounds also contain hydrogen. Some of the most important compounds contain elements such as nitrogen and oxygen as well. In this topic, we look at some of the large numbers of compounds that contain only carbon and hydrogen. These compounds are called **hydrocarbons**.

If an organic compound contains other elements as well as carbon and hydrogen, then it is not a hydrocarbon. For example, many foods contain a sugar called sucrose. Sucrose contains carbon and hydrogen, but also oxygen, so it is not a hydrocarbon. It is an example of a carbohydrate – the *-ate* ending shows that it contains oxygen.

DID YOU KNOW?

In the 1960s, the organic chemist Stephanie Kwolek invented an extremely strong polymer called Kevlar®. Kevlar fibre is now widely used in many applications, including heat resistant gloves, smartphones, sports equipment and body armour.



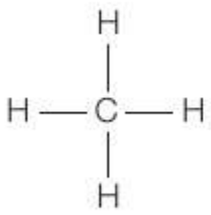
▲ Award-winning US chemist Stephanie Louise Kwolek was inducted into the National Inventors Hall of Fame in 1995.

SATURATED OR UNSATURATED?

Although there are many thousands of different hydrocarbons, most of them are classed as **saturated** or **unsaturated**. Like many chemical terms, these words have a very different meaning in everyday life. Someone who has been caught in a heavy rain shower may say that their clothes are saturated, which means that they have absorbed as much water as they possibly can.

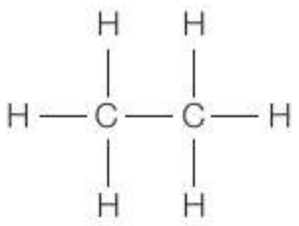
In organic chemistry, these terms have nothing to do with water, although there is a connection. A hydrocarbon that is saturated contains as much hydrogen as possible, which depends on the number of carbon atoms in the molecule. If a hydrocarbon has fewer hydrogen atoms than the maximum, then it is not saturated, we say it is unsaturated.

The formula of the simplest hydrocarbon, containing only one carbon atom, is:

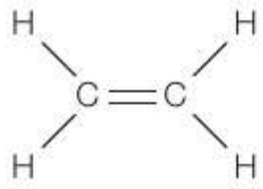


When a hydrocarbon contains two carbon atoms, there is a maximum of six hydrogen atoms.

There is a hydrocarbon that contains two carbon atoms and six hydrogen atoms, but also one that contains two carbon atoms and only four hydrogen atoms. The formulae of these hydrocarbons are:



saturated



unsaturated

You can see that in all three examples, each carbon atom has four bonds to other atoms. This is a general rule for organic compounds. In most cases, every carbon atom has four bonds.

The difference between a saturated hydrocarbon and an unsaturated hydrocarbon is to do with whether or not there is room, for more hydrogen atoms.

- If there is no room, then the hydrocarbon is saturated.
- If there is room, then the hydrocarbon is unsaturated.

One easy way to decide whether a hydrocarbon is saturated or unsaturated is to look at structures like the ones above.

- If there are two bonds (a double bond) drawn between one or more carbon atoms, then the hydrocarbon is unsaturated.
- If there are only single bonds, then the hydrocarbon is saturated.

ALKANES AND CYCLOALKANES

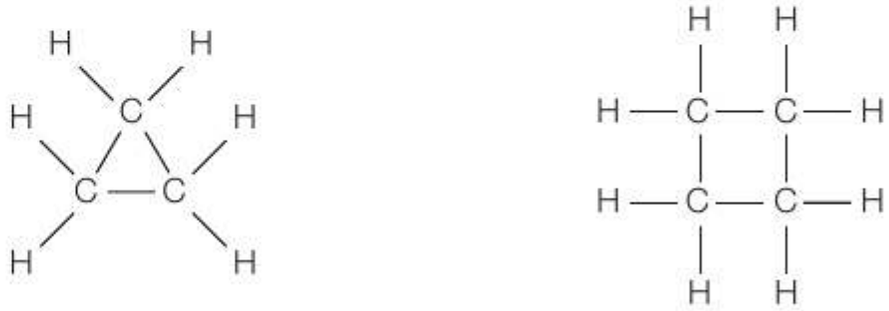
Look at the formulae in **fig B**. These are alkanes.



▲ **fig B** Propane and butane.

Both structures have only single bonds and so are saturated.

Now look at the formulae in **fig C**. These are cycloalkanes.



▲ **fig C** Cyclopropane and cyclobutane.

Both structures have only single bonds and so are saturated. They both have carbon atoms joined in a ring structure. The first has a triangular arrangement and the second has a square arrangement. These ring structures mean that they are cyclic compounds. We will look at alkanes and cycloalkanes in more detail in **Topic 4B**.

EXAM HINT

It is important to remember that these structures are actually three dimensional. The bond angle in cyclobutane looks like it is 90 degrees as it is drawn in 2D but in fact is closer to 109.5 degrees as it would be in methane.

LEARNING TIP

The formulae of the hydrocarbons in this topic show every bond and every atom. These formulae can also be written as molecular formulae. The first one can be written as CH_4. Try writing molecular formulae for the other hydrocarbons shown in this topic.

CHECKPOINT

1. A saturated hydrocarbon contains five carbon atoms in its molecule. What are the two possible molecular formulae for this hydrocarbon?

2. Consider the compound with the formula $C_2H_4O_2$. Is it an organic compound? Explain your answer.

SUBJECT VOCABULARY

hydrocarbon a compound that contains only carbon and hydrogen atoms
saturated a compound containing only single bonds
unsaturated a compound containing one or more double bonds

4A 2 DIFFERENT TYPES OF FORMULAE

SPECIFICATION REFERENCE 4.5(ii)

LEARNING OBJECTIVES

- Draw compounds and represent organic molecules using structural, displayed and skeletal formulae.

EXAM HINT

If an exam question asks for a particular type of formula in an answer, it is important that you use it. If it asks for a displayed formula, then a skeletal formula will not gain full marks.

USING DIAGRAMS TO REFER TO ORGANIC COMPOUNDS

There are many millions of organic compounds, so making clear which ones we are referring to can be challenging.

There are two main ways to refer to organic compounds. We can use:

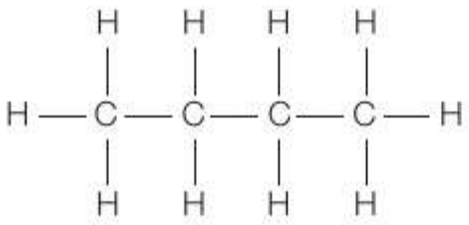
- formulae
- names.

In this topic we will consider how to refer to organic compounds using formulae.

DISPLAYED FORMULAE

The formulae you saw in **Topic 4A.1** are all **displayed formulae**. They show (display) every atom and every bond separately. In many situations, these are the best type of formulae to use, but sometimes it is better to simplify them.

Consider the hydrocarbon with this displayed formula, its name is butane:



STRUCTURAL FORMULAE

One way to simplify this displayed formula is to group all the atoms joined to a particular carbon atom together. We can choose to show the bonds between the carbons, or we can leave them out. These are both **structural formulae** of butane:

CH_3— CH_2— CH_2— CH_3 and $CH_3CH_2CH_2CH_3$

SKELETAL FORMULAE

Another way to represent a compound is by a **skeletal formula**. The word skeletal is connected with the word skeleton, which, as you know, shows only the bones in a human or animal body.

A skeletal formula is a zig-zag line that shows only the bonds between the carbon atoms. Every change in direction and every ending means that there is a carbon atom (with as many hydrogen atoms as needed). Atoms other than carbon and hydrogen need to be shown.

This is the skeletal formula of butane:



The start and end both represent CH_3, and the two junctions between lines each represent CH_2.

MOLECULAR FORMULAE

The displayed, structural and skeletal formulae above show the structures of the molecules unambiguously. In other words, each formula represents only one compound. With a displayed formula, this is very clear. With a structural formula, you have to imagine how the atoms are joined together in groups such as CH_2 and CH_3, but that is very straightforward. With a skeletal formula, once you know the rules, you can be sure how every atom is arranged in the molecule.

Now consider the formula C_3H_7Cl. This is clearly not a displayed formula or a skeletal formula, but it is also not a structural formula. This is because, with three carbon atoms, the chlorine atom could be attached to the middle carbon atom or to either of the end carbon atoms. The formula C_3H_7Cl actually represents two different compounds.

Formulae like these are called **molecular formulae.** They only show the numbers of each type of atom in the molecule, and not its structure. Of course, in very simple molecules such as CH_3Cl, the molecular formula can be used to work out the displayed, structural and skeletal formulae because there is only one way in which these five atoms can be joined together.

EMPIRICAL FORMULAE

Another type of formula is an **empirical formula**. This shows the compound like a molecular formula, but the numbers of each atom are in their simplest possible whole-number ratio. This means that butane (molecular formula C_4H_{10}) has an empirical formula of C_2H_5.

In chemistry, the word empirical usually means 'as found from practical evidence'. You would normally work out this type of formula mathematically from the results of an experiment. You can see how we do this in **Topic 1D.1**.

DIFFERENT TYPES OF FORMULA FOR CHLOROETHANE

Until now, we have only considered the different types of formula using a hydrocarbon as the example.

Consider an example containing a third element (chlorine) – chloroethane. **Table A** shows the different types of formula for chloroethane.

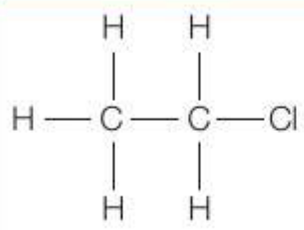
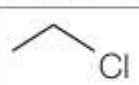
TYPE OF FORMULA	FORMULA
displayed formula	
structural formula	$CH_3–CH_2–Cl$ or CH_3CH_2Cl
skeletal formula	
molecular formula	C_2H_5Cl
empirical formula	C_2H_5Cl

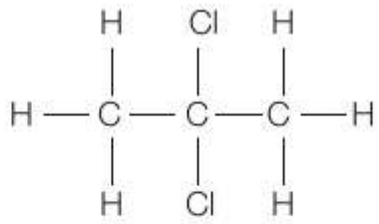
table A Different types of formula for chloroethane.

LEARNING TIP

For some compounds, the numbers in the molecular formula cannot be simplified. This means that the molecular formula and the empirical formula are identical.

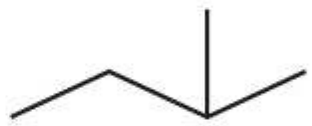
CHECKPOINT

1. The displayed formula of a compound is:



Use a table (like the one for chloroethane) to show the structural, skeletal, molecular and empirical formulae of the compound above.

2. The skeletal formula of a compound is:



Use a table (like the one for chloroethane) to show the displayed, structural, molecular and empirical formulae of the compound above.

SUBJECT VOCABULARY

displayed formula shows every atom and every bond

structural formula shows (unambiguously) how the atoms are joined together

skeletal formula shows all the bonds between carbon atoms

molecular formula shows the actual numbers of each atom in the molecule

empirical formula shows the numbers of each atom in the simplest whole-number ratio

4A 3 FUNCTIONAL GROUPS AND HOMOLOGOUS SERIES

SPECIFICATION REFERENCE 4.4

LEARNING OBJECTIVES

- Understand the concepts of homologous series and functional groups.

FUNCTIONAL GROUP

A **functional group** in a molecule is an atom or group of atoms that gives the compound some distinctive and predictable properties. For example, the functional group of atoms shown as COOH gives substances such as vinegar a sour, acidic taste.

There are many organic compounds containing this group. Here are some examples:

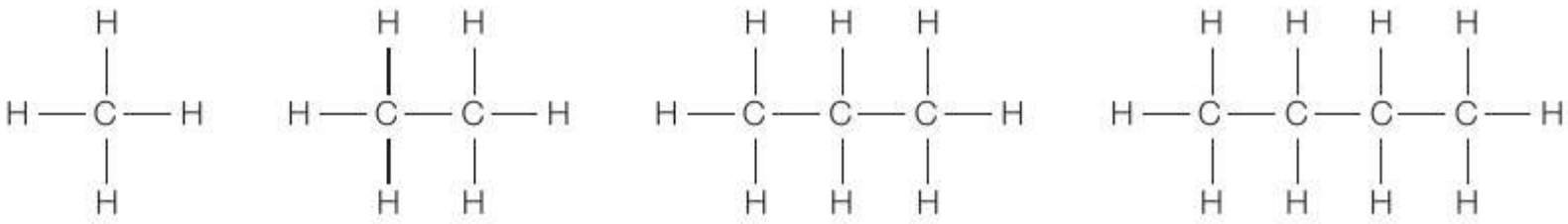
$HCOOH$ CH_3COOH CH_3CH_2COOH $CH_3CH_2CH_2COOH$

HOMOLOGOUS SERIES

If you look at the formulae above, you can see that each formula has one more carbon atom and two more hydrogen atoms than the previous one, they differ by CH_2. These compounds are the first four members of what is called a **homologous series**. A homologous series is a set of compounds with the same functional group, similar chemical properties, and physical properties that show a gradation (a gradual change from one to the next).

ALKANES

The organic compounds that are mainly used as fuels are the alkanes (you will learn more about alkanes in **Topic 4B**). Alkanes are not considered to contain a functional group, but otherwise they form a homologous series. The displayed formulae of some alkanes are:



GENERAL FORMULAE

In **Topic 4A.2,** we looked at five different types of formulae. Now we are going to look at another type of formula. For the compounds in a homologous series, we can use a general formula to represent all of them. This is done by using the letter *n* for the number of carbon atoms, excluding any in the functional group. For the compounds with formulae ending in COOH, the general formula is $C_nH_{2n+1}COOH$.

Table A shows the formulae for some of the homologous series in this book.

NAME	GENERAL FORMULA	EXAMPLE
alkane	C_nH_{2n+2}	CH_4
alkene	C_nH_{2n}	C_2H_4
halogenoalkane	$C_nH_{2n+1}X$	CH_3CH_2Br
alcohol	$C_nH_{2n+1}OH$	CH_3CH_2OH

table A Examples of homologous series used in this book.

PROPERTIES OF A HOMOLOGOUS SERIES

ALKANES

We can use the alkanes to illustrate the similarity in chemical properties of a homologous series. For example, when alkanes are burned completely in air, they all form the same two products: carbon dioxide and water.

The commonest alkane is methane. The equation for the complete combustion of methane is:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

ALCOHOLS

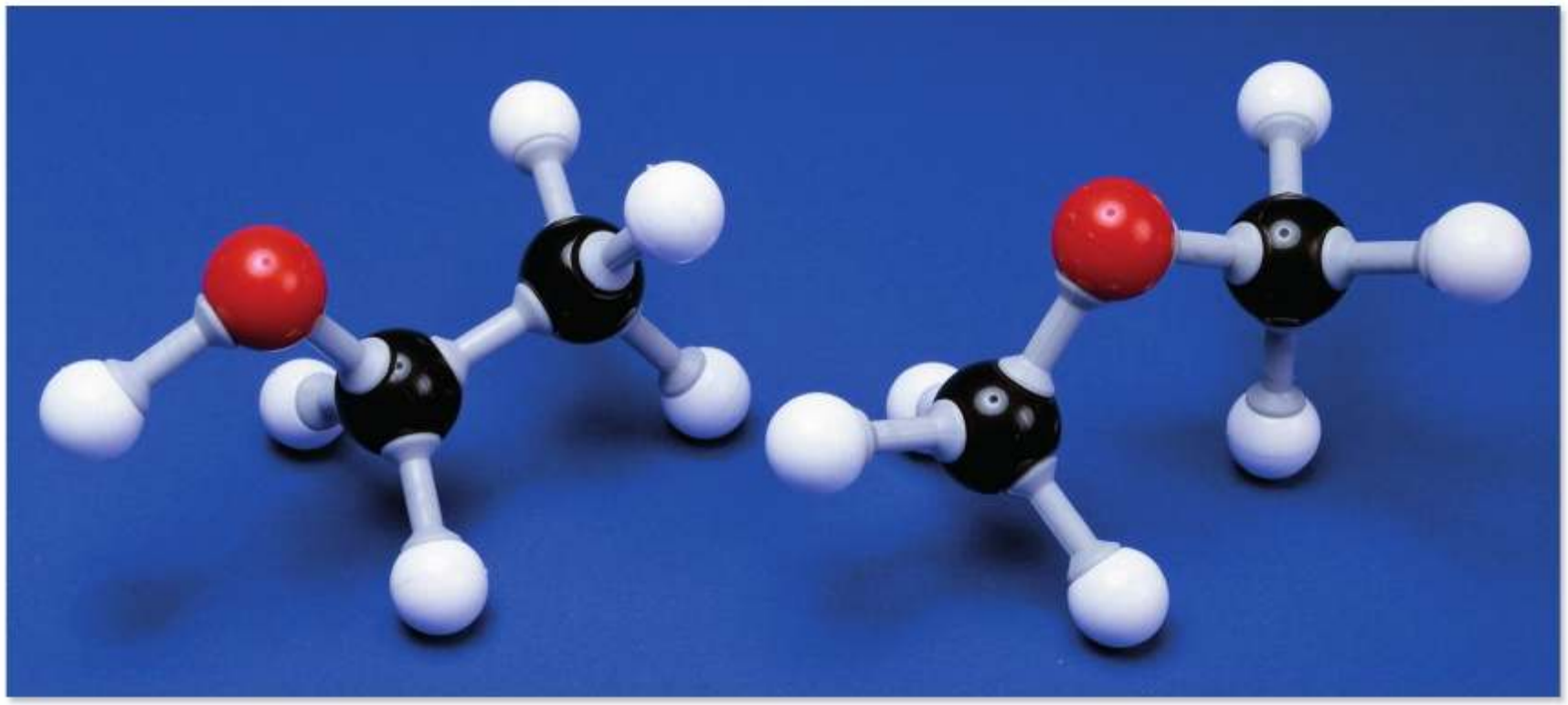
We can use the alcohols to illustrate the gradation in physical properties of a homologous series. For example, the boiling temperatures of the first four alcohols are shown in **table B**.

FORMULA	BOILING TEMPERATURE / °C
CH_3OH	65
CH_3CH_2OH	79
$CH_3CH_2CH_2OH$	97
$CH_3CH_2CH_2CH_2OH$	117

table B You can see that as the number of carbon and hydrogen atoms increases, so does the boiling temperature.

EXAM HINT

An exam question may ask you to plot the boiling point of organic compounds (such as alcohols) with carbon number on the *x*-axis. Be sure to use a suitable scale and at least half of the axes provided in both directions.



▲ **fig A** Molecular models are very useful in organic chemistry. Both of these structures contain an oxygen atom (shown in red), but you can see that they belong to different homologous series.

LEARNING TIP

When looking at molecular models, remember that different elements are represented by different colours. The most common colours are black for carbon, white for hydrogen and red for oxygen.

CHECKPOINT

1. The equation for the complete combustion of propane is:

$$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$$

What is the equation for the complete combustion of the alkane with five carbon atoms in its molecule?

2. The structural formula of a compound is CH_3CH_2CHO. What are the formulae of the two simpler compounds in the same homologous series?

SUBJECT VOCABULARY

functional group an atom or group of atoms in a molecule that is responsible for its chemical reactions

homologous series a family of compounds with the same functional group, which differ in formula by CH_2 from the next member

4A 4 NOMENCLATURE

SPECIFICATION REFERENCE
4.5(i)

LEARNING OBJECTIVES

- Apply the rules of IUPAC nomenclature to name compounds relevant to this course.

WHY DO WE NEED RULES FOR NAMING ORGANIC COMPOUNDS?

As the number of known organic compounds has increased, it has become harder to continue to find new names for them. In **Topic 4A.3**, we referred to the simplest organic compound (CH_4) as methane, but it was originally known as marsh gas (because it was found in marshes, where it was formed by the decay of plants). Many other organic compounds were named in similar ways.

An organisation called the International Union of Pure and Applied Chemistry (usually abbreviated to IUPAC ('eye-you-pack')) made some rules about how to name organic compounds. These rules are known as 'nomenclature'. The detailed rules needed for naming very complicated compounds are complex, but the simpler rules for the compounds described in your IAS course are much easier to understand and apply.

THE SIMPLE RULES OF NOMENCLATURE

Table A summarises the principles of naming organic compounds, including rules for **prefixes**, **suffixes** and **locants**.

THE PART OF THE NAME	HOW TO WRITE IT	EXAMPLE
Number of carbon atoms	This is shown by using a letter code (usually three or four letters).	meth = one carbon atom
Prefixes Suffixes	The presence of atoms other than carbon and hydrogen is shown by adding other letters before or after the code for the number of carbon atoms.	bromo = an atom of bromine ol = a hydroxyl group (OH)
Multiplying prefixes	The presence of two or more identical groups is shown by using the prefixes di-, tri-, etc.	di = two
Locants	Where atoms and groups can have different positions in a molecule, numbers and hyphens are used to show their positions. The numbers represent the carbon atoms in the longest chain that the atoms and groups are attached to.	2- = the atom or group is attached to the second carbon atom in the chain

table A The principles of naming organic compounds.

The letter codes for the number of carbon atoms (up to ten) are shown in **table B**.

NUMBER	CODE	PREFIX
1	meth	methyl
2	eth	ethyl
3	prop	propyl
4	but	butyl
5	pent	pentyl
6	hex	hexyl
7	hept	heptyl
8	oct	octyl
9	non	nonyl
10	dec	decyl

table B

APPLYING THE RULES TO WRITE NAMES

ALKANES

We can see how these rules work for some of the alkanes (**table C**). The names of all the alkanes end in *-ane*.

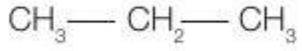
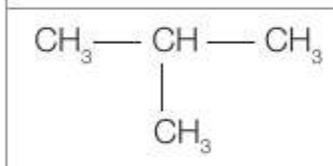
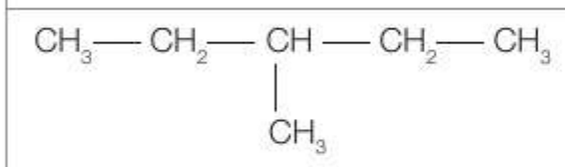
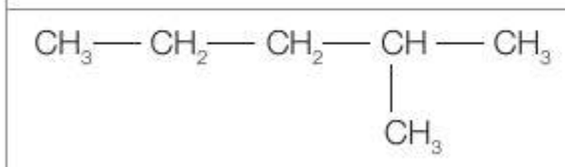
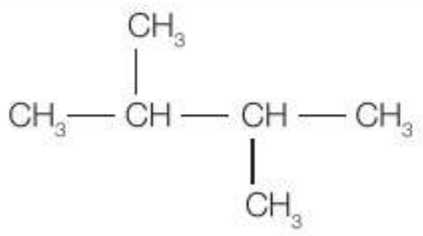
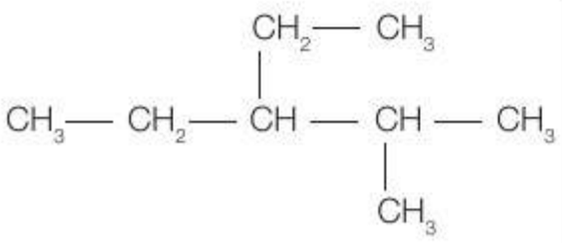
STRUCTURAL FORMULA	NAME
	propane
	methylpropane The locant 2- is not needed because if the methyl group below the horizontal chain were attached to one of the carbon atoms at either end of the chain, then there would be a sequence of four carbon atoms, and the compound would be named butane.
	3-methylpentane The longest carbon chain contains five carbon atoms, and there is a methyl group attached to the third one.
	2-methylpentane This is not 4-methylpentane because another rule is that the lowest locant numbers should be used.
	2,3-dimethylbutane This example shows the use of a comma between the locants when the attached groups are the same.
	3-ethyl-2-methylpentane This example illustrates the rule about prefixes being in alphabetical order. Ethyl comes before methyl because e comes before m in the alphabet. Notice also that it is not called 3-ethyl-4-methylpentane because these numbers (3 + 4) total more than the numbers 3 + 2 in the correct name.

table C Naming alkanes from structural formulae using the rules of IUPAC nomenclature.

ALCOHOLS

Next, look at the alcohols in **table D**. The rules for these are a bit different because the presence of the alcohol functional group is indicated by a suffix, not a prefix. The names for all the alcohols end in *-ol*.

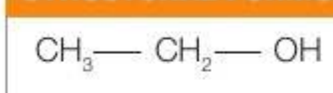
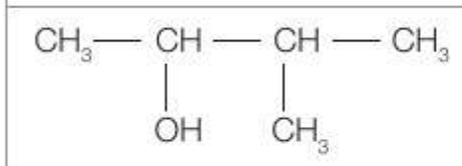
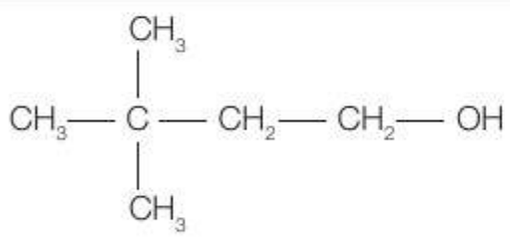
STRUCTURAL FORMULA	NAME
	ethanol
	propan-1-ol This time, the locant appears near the end of the name, but, as before, it appears directly before the letters representing the group.
	3-methylbutan-2-ol This example illustrates the use of both a prefix and a suffix. This is not called 2-methylbutan-3-ol because the lowest number locant should be used for the suffix functional group (-2-ol not -3-ol).
	3,3-dimethylbutan-1-ol This example shows the use of prefixes, a suffix, locants and a comma. As with alkanes, the name uses locants that add up to the smallest possible number.

table D Naming alcohols from structural formulae using the rules of IUPAC nomenclature.

APPLYING THE RULES TO WRITE FORMULAE

Table E gives some examples of applying the rules the other way round, i.e. writing a structural formula for a compound from its IUPAC name.

LEARNING TIP

When you practise writing names from structural formulae, always check that the code you have used for the longest carbon chain is for the longest chain. This may not be the one shown horizontally. When you practise writing structural formulae from names, always check that each carbon has only four bonds. Showing three or five bonds is a common error.

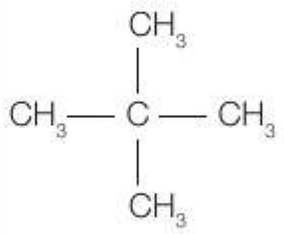
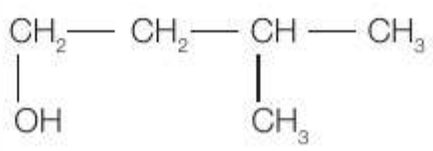
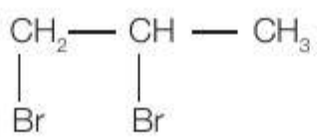
NAME	STRUCTURAL FORMULA
dimethylpropane	*prop* indicates a chain of three carbon atoms *dimethyl* indicates two methyl groups attached to the chain No locants are used, so the two methyl groups must be attached to the carbon chain in a way that does not make the longest carbon chain any longer than three carbon atoms. So the structural formula is: 
3-methylbutan-1-ol	*but* indicates a chain of four carbon atoms *methyl* indicates a CH_3 group *1-* and *3-* indicate attachments to the first and third carbon atoms in the chain So the structural formula is: 

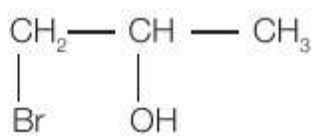
table E Writing structural formulae from IUPAC names.

CHECKPOINT

SKILLS REASONING

1. Write IUPAC names for the compounds with these structural formulae.

(a) 

(b) 

2. Write structural formulae for the compounds with these IUPAC names.

(a) 2,2-dimethylpentane (b) 2,3-dimethylbutan-2-ol

SUBJECT VOCABULARY

prefix a set of letters written at the beginning of a name
suffix a set of letters written at the end of a name
locant a number used to indicate which carbon atom in the chain an atom or group is attached to

4A 5 STRUCTURAL ISOMERISM

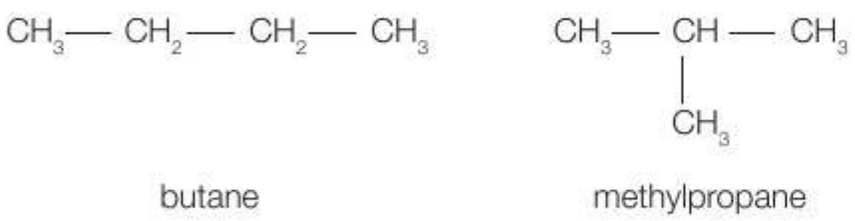
SPECIFICATION REFERENCE
4.10 4.11

LEARNING OBJECTIVES

- Understand the term 'structural isomerism' and draw the structural isomers of organic molecules, given their molecular formula.
- Draw the structural isomers of alkanes and cycloalkanes with up to six carbon atoms.

STRUCTURAL ISOMERS

Consider these two structures:



butane methylpropane

You can see that they are different compounds because their names and structures are different. However, their molecular formulae are the same. They can both be represented by C_4H_{10}. These two compounds are simple examples of **structural isomers**. In other words, they have the same molecular formula but different structural formulae.



▲ **fig A** By counting the atoms you can see that both structures have the molecular formula C_4H_{10}.

The molecular formula C_4H_{10} represents only two possible structures, but more complicated molecular formulae can be represented by several possible structures. As the number of carbon atoms increases, the number of possible structures can be hundreds or thousands.

TYPES OF STRUCTURAL ISOMERISM

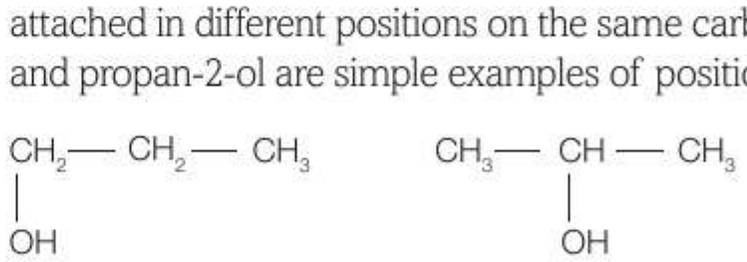
There are two important types of structural isomerism.

CHAIN ISOMERISM

Chain isomerism refers to molecules with different carbon chains. Butane and methylpropane (shown in **fig A** above) are examples of chain isomers because their carbon chains are different.

POSITION ISOMERISM

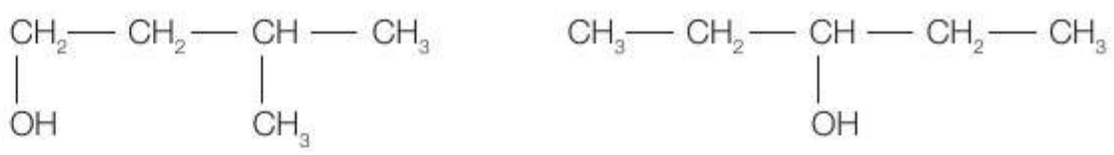
Position isomerism refers to molecules with the same functional group attached in different positions on the same carbon chain. Propan-1-ol and propan-2-ol are simple examples of position isomerism:



propan-1-ol propan-2-ol

They are examples of position isomers because the carbon chains are the same, but the OH groups are attached to different carbon atoms in the chain.

You might see examples where both of these types of isomerism are present:



DRAWING STRUCTURAL ISOMERS FROM MOLECULAR FORMULAE

You will remember that a molecular formula shows the actual numbers of atoms of each element, but does not show how the atoms are arranged. In some cases (such as C_2H_6) there is only one possible structure that can be drawn. In many cases there are lots of structures that can be drawn.

When you draw a structure, you could use a displayed formula, but with many bonds and atoms the formula would look complicated. At this stage, it is best to draw structural formulae. These formulae show every carbon (and other atoms such as oxygen) separately, with bonds between these atoms. The hydrogen atoms are grouped together to keep the structure simpler.

A good way to begin is by showing all the carbon atoms separately, in a straight line and with bonds between them. Then show the other atoms in as many different ways as possible, starting with atoms other than hydrogen.

Next see if there is another way to show the carbon atoms, not as a straight chain but as a shorter chain with one or more branches. Then add the other atoms in as many different ways as possible.

Time for some examples.

It is useful to refer to each carbon in a chain as C1, C2, etc.

WORKED EXAMPLE

Draw the structural isomers for the molecular formula $C_4H_8Cl_2$.

Step 1 The four carbon atoms can all be in a straight line, or in a straight line of three with a branch from the middle carbon to the fourth carbon. Let's start with the straight line of four:
C– C– C– C

Step 2 There are two Cl atoms. They could both be attached to C1 or to C2 in this chain. You might think that they could be attached to C3 and C4 as well. They could be, but counting from the right, C1 and C4 are the same, and C2 and C3 are the same.

The two Cl atoms could also be attached to different carbon atoms, such as C1 and C2, or C2 and C3. It will take a lot of practice before you decide on the correct number of different structures. You might miss some possible structures, or write two structures that look different but are actually the same.

Step 3 Finally, add the necessary hydrogen atoms, making sure that every carbon atom has four bonds.

Table A shows the approach.

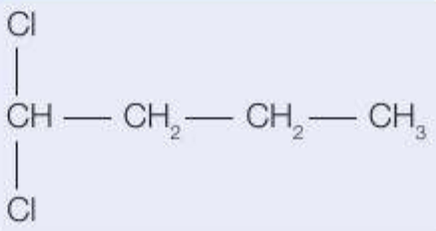
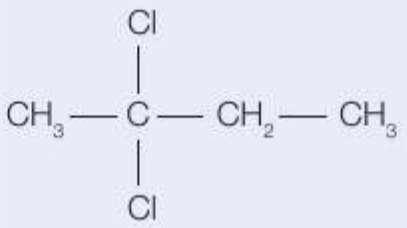
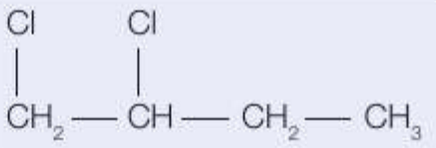
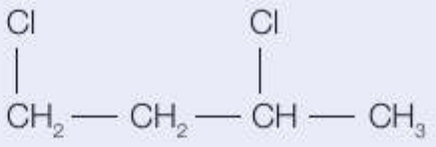
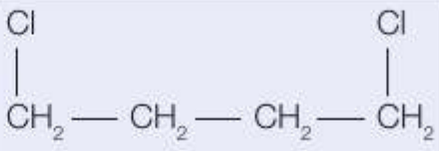
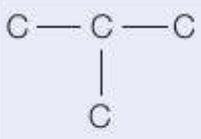
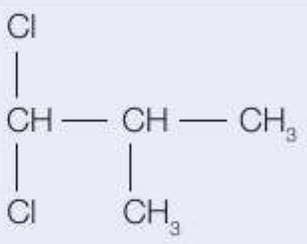
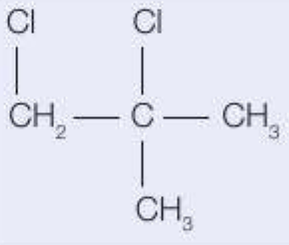
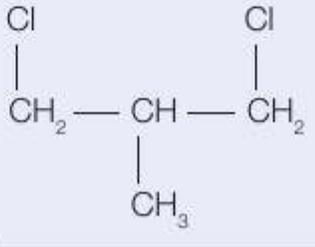
CARBON CHAIN	ATTACHING THE TWO Cl ATOMS	RESULT
C– C– C– C	Both on C1	
	Both on C2	
	On C1 and C2	
	On C1 and C3	
	On C1 and C4	
	Both on C1	
	On C1 and C2	
	On C1 and C3	

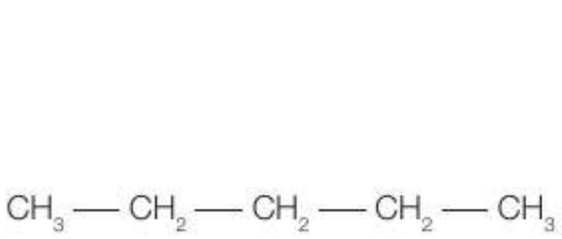
table A

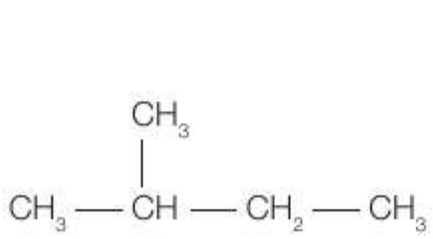
You may think there are other possible structures. These others may look different on paper, but careful checking will show that they have already been used. Here is a useful way to be sure whether two structures are different, or whether they look different but are actually the same. Apply the IUPAC rules to name the structures. If this gives you two completely different names, then the structures are different. If it gives you the same names, then the structures are the same.

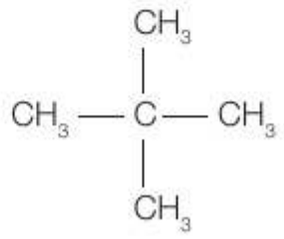
ALKANES AND CYCLOALKANES

You need to be able to draw the structures of all the alkanes and cycloalkanes with up to six carbon atoms. You have already seen structures of some examples of these homologous series, but now let us adapt the method we used for the structures of $C_4H_8Cl_2$. We can do this for the compounds with five carbon atoms as examples.

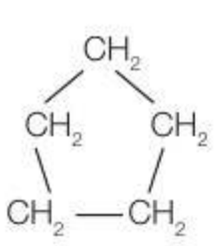
First, the alkanes. With 5 carbon atoms, there are only three different ways to arrange the carbon atoms. These are: (1) a line of 5; (2) a line of 4 with the 5th carbon branching off one of the central carbons; and (3) a line of 3 with the 4th and 5th carbons both branching off the middle one. Once you have realised this, you can just write these three possibilities, making sure that each carbon has the correct number of hydrogen atoms. Remember that each carbon atom must have 4 bonds, and the total number of hydrogen atoms must be 12.

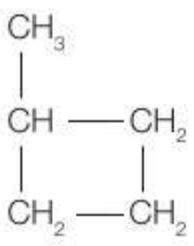


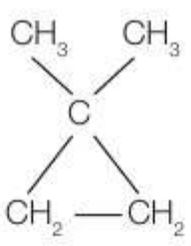


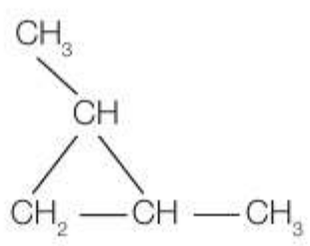


Now for the cycloalkanes. There can be a ring of 5 carbon atoms; or a ring of 4 carbon atoms with the 5th carbon atom attached to any one of the 4; or a ring of 3 carbon atoms with the 4th and 5th attached to the same or different carbons in the ring. Remember that each carbon atom must have 4 bonds, and the total number of hydrogen atoms must be 10. These are the possibilities.









LEARNING TIP

Try drawing structures for all the possible isomers of the alkanes with six carbon atoms in the molecule. You will probably draw more than there really are. You can then decide which ones you don't need by naming them. If they look different but have the same name, then they are the same.

CHECKPOINT

1. Draw a structure for each of the alkane isomers with the molecular formula $C_3H_6Br_2$.

2. Draw a structure for each of the cycloalkane isomers with the molecular formula $C_3H_4Br_2$.

SUBJECT VOCABULARY

structural isomers compounds with the same molecular formula but different structural formulae

4A 6 TYPES OF REACTION

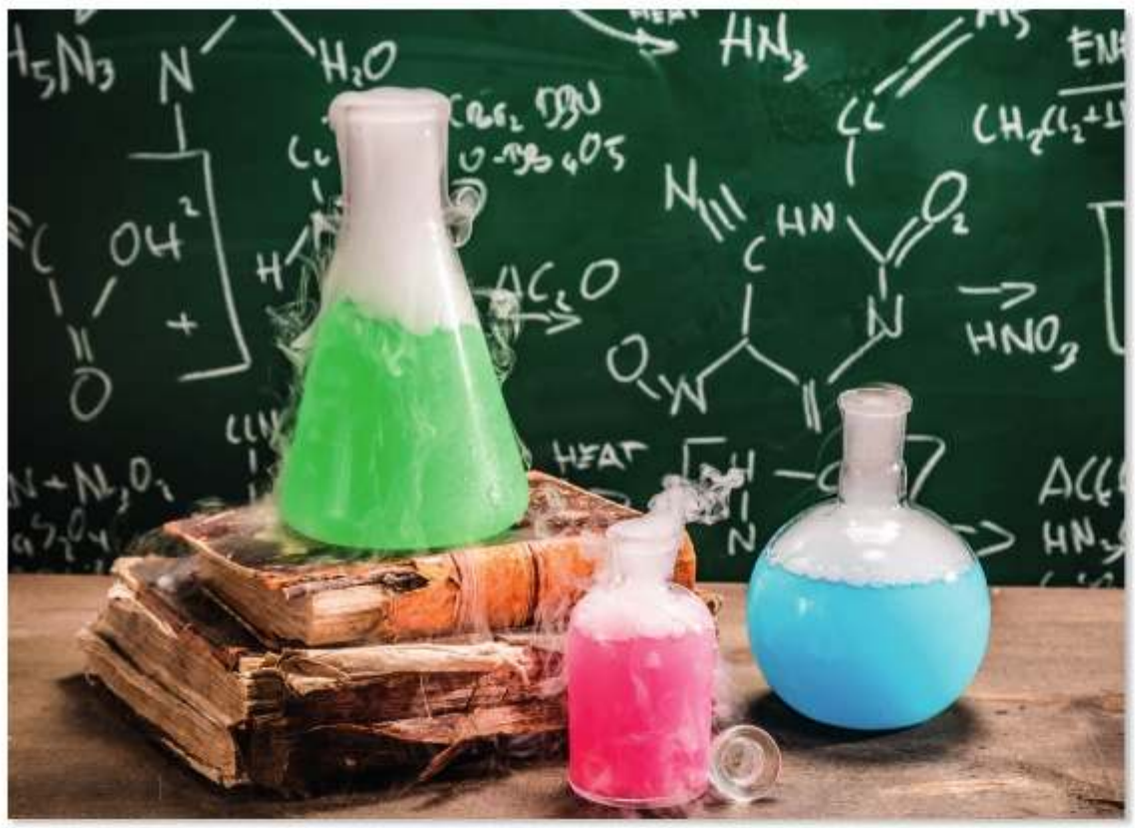
SPECIFICATION REFERENCE: 4.6 4.7 4.8

LEARNING OBJECTIVES

- Classify reactions as addition, substitution, oxidation, reduction or polymerisation.
- Understand that bond breaking can be homolytic (to produce free radicals) or heterolytic (to produce ions).
- Know the definitions of the terms 'free radical' and 'electrophile'.

REACTIONS IN ORGANIC CHEMISTRY

In later topics of this book, you will learn about many different reactions and see many equations to represent these reactions. It will help if you can recognise these reactions as belonging to one of five main types. There is much more detail about these reactions in later topics (for example, **Topic 10**).



▲ **fig A** What types of reaction are occurring in these flasks?

ADDITION REACTIONS

In an **addition reaction**, two reactant species combine together to form a single product species. Usually all the species are molecules. A general equation for this type of reaction is:

$$A + B \rightarrow C$$

One example is the reaction between ethene and bromine:

$$C_2H_4 + Br_2 \rightarrow C_2H_4Br_2$$

SUBSTITUTION REACTIONS

In a **substitution reaction**, two reactant species combine together to form two product species. Usually all the species are molecules or ions. A general equation for this type of reaction is:

$$A + B \rightarrow C + D$$

One example is the reaction between bromoethane and potassium hydroxide. Potassium hydroxide is an ionic compound, and as the potassium ion is a spectator ion, the reaction only involves the hydroxide ion. An equation for this reaction is:

$$C_2H_5Br + OH^- \rightarrow C_2H_5OH + Br^-$$

In this reaction, the OH group has taken the place of, or *substituted*, the Br atom.

OXIDATION REACTIONS

In an **oxidation reaction**, one organic compound is oxidised, usually by an inorganic reagent. This means that the organic compound can either lose hydrogen or gain oxygen. There is not a suitable general equation that can be used for this type of reaction, but here is one example you will find in **Topic 10** – the oxidation of ethanol by a mixture of potassium dichromate(VI) and sulfuric acid. The equation is not written to include the inorganic reagent as it would be very complicated. Usually the oxygen atoms produced by the oxidising agent are shown using the symbol [O], so the equation then becomes:

$$C_2H_5OH + [O] \rightarrow CH_3CHO + H_2O$$

This reaction is classified as oxidation because the ethanol molecule loses two hydrogen atoms.

EXAM HINT

It is important that you use the symbol [O] for an oxidising agent. O_2 would be incorrect in this case as it suggests that the oxidising agent is molecular oxygen from the air.

REDUCTION REACTIONS

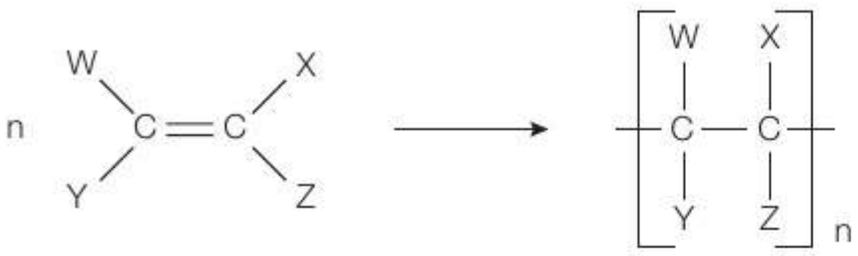
In a **reduction reaction**, one organic compound is reduced, sometimes by hydrogen gas and a catalyst and sometimes by an inorganic reagent. This means that the organic compound can either gain hydrogen or lose oxygen. There is not a suitable general equation that can be used for this type of reaction, but here is one example you will find in **Topic 5A.3**. This is the reduction of an alkene to an alkane by hydrogen gas and a nickel catalyst. The equation for the reaction is:

$$C_2H_4 + H_2 \rightarrow C_2H_6$$

You can now see why this reaction is classified as reduction. The ethene molecule gains two hydrogen atoms. Note that this is also an example of an addition reaction.

POLYMERISATION REACTIONS

In this book, all the **polymerisation reactions** you will meet are examples of addition polymerisation. In addition polymerisation, very large numbers of a reactant molecule (sometimes of two different reactant molecules) react together to form one very large product molecule. A general equation for this type of reaction is:



A familiar example of this type of reaction is the polymerisation of ethene to poly(ethene).

BOND BREAKING IN ORGANIC REACTIONS

Organic compounds contain covalent bonds, for example between

- two carbon atoms
- a carbon atom and a hydrogen atom
- a carbon atom and a halogen atom.

There are two different ways for the covalent bond to break. These are **homolytic fission** and **heterolytic fission**.

HOMOLYTIC FISSION

Like many other scientific terms, 'homolytic' comes from Greek: 'homo' indicates 'same' and 'lytic' indicates 'splitting'.

In homolytic fission, the shared pair of electrons in the covalent bond divide equally between the two atoms. This can be shown like this:



Each product species keeps one of the electrons from the covalent bond. These species are called **free radicals**. Each free radical has an unpaired electron and is uncharged.

Homolytic fission usually occurs when the two atoms bonded together are identical or when they have similar electronegativities.

HETEROLYTIC FISSION

'Heterolytic' is another term from Greek: 'hetero' indicates 'different' and 'lytic' indicates 'splitting'.

In heterolytic fission, both electrons of the shared pair in the covalent bond are kept by one of the atoms. This can be shown like this:



The left-hand product species keeps both of the electrons from the covalent bond. This species is a negative ion. The right-hand product species does not keep either of the electrons from the covalent bond. This species is a positive ion.

Heterolytic fission could also occur like this:



Here there are still two ions formed, but this time the right-hand product is the negative ion.

Heterolytic fission usually occurs when the two atoms bonded together have different electronegativities. The atom with the higher electronegativity is the one that keeps both electrons from the bond.

ELECTROPHILES

The origin of the term **electrophile** is 'electron', which indicates negative charge, and 'phile', which means liking. You may have come across the word 'bibliophile', which means a person who likes books. An electrophile refers to a chemical species that 'likes negative charge'. So how does this term fit in with this topic? You remember that opposite charges (positive and negative) attract each other. The positive ion produced by heterolytic fission will be attracted to a region of high electron density in another molecule. This region is often labelled with the symbol $\delta-$. You will learn much more about electrophiles and their importance in reaction mechanisms in **Topic 5A.4**.

LEARNING TIP

Look at some equations in **Topic 4** and try to classify at least two as addition, substitution, oxidation, reduction or polymerisation.

CHECKPOINT

1. What type of reaction is shown by each of the following equations?

(a) $CH_3CHO + 2[H] \rightarrow CH_3CH_2OH$

(b) $CH_3CHO + [O] \rightarrow CH_3COOH$

(c) $CH_3CHO + HCN \rightarrow CH_3CH(OH)CN$

(d) $CH_3Br + KOH \rightarrow CH_3OH + KBr$

2. Write an equation to represent each of the following.

(a) Homolytic fission of the carbon-to-carbon bond in C_2H_6.

(b) Heterolytic fission of the carbon-to-bromine bond in C_2H_5Br.

SUBJECT VOCABULARY

addition reaction reaction in which two molecules combine to form one molecule

substitution reaction reaction in which one atom or group is replaced by another atom or group

oxidation reaction reaction in which a substance gains oxygen or loses hydrogen

reduction reaction reaction in which a substance loses oxygen or gains hydrogen

polymerisation reaction reaction in which a large number of small molecules react together to form one very large molecule

homolytic fission the breaking of a covalent bond where each of the bonding electrons leaves with one species, forming a free radical

heterolytic fission the breaking of a covalent bond so that both bonding electrons are taken by one atom

free radical a species that contains an unpaired electron

electrophile a species that is attracted to a region of high electron density

4A 7 HAZARDS, RISKS AND RISK ASSESSMENTS

SPECIFICATION REFERENCE
4.1 4.2 4.3

LEARNING OBJECTIVES

- Understand the difference between hazard and risk.
- Understand the hazards associated with organic compounds and why it is necessary to carry out risk assessments when dealing with potentially hazardous materials.
- Suggest ways in which risks can be reduced and reactions carried out safely.

SAFETY IN CHEMISTRY LABORATORIES

Incidents that cause harm to people are rare in school and college laboratories. One of the reasons for this is that laboratories need to consider the hazards of doing chemistry experiments and use safe methods of working.

This applies to all chemistry experiments, but especially to those involving organic compounds. It is particularly important in experiments that you plan yourself, but also for those where you are following a method you have been given.

When you plan an organic synthesis, you need to consider the hazards associated with the reactants, the substance you are synthesising, and also any intermediates formed.

HAZARD AND RISK

The **hazard** of a chemical substance relates to the inherent properties of the substance. The **risk** is more to do with how you plan to use it and the chance of it causing harm. Most people would consider that water is completely safe and has no hazards. In most situations, this is the case. However, consider a beaker of water being boiled on a tripod and gauze. The steam coming from the water, and the boiling water itself, could both cause harm if they came into contact with your skin.

Now consider a substance that most people would consider hazardous. You have probably used hydrochloric acid in experiments involving marble chips. You will have been told to use eye protection when using the acid because of the harm it could do, especially if it got into your eyes.

The hazard exists because hydrochloric acid is corrosive and so could damage your eyes..

The risk depends on how likely it is that the acid can get into your eyes.

Using eye protection does not affect the hazard, but greatly decreases the risk.

HAZARD WARNING SYMBOLS

A long time ago, bottles containing certain chemicals were labelled with the word POISON. This early attempt to prevent harm to laboratory workers was well-intentioned, but some people might think that bottles without such a label contained harmless substances.

More recently, symbols (sometimes called pictograms) have been used to label substances but also to identify the actual hazard of the substance inside. The actual symbols have changed over the years, and you may see older and newer types together in the same laboratory.

Older ones are often square in shape with an orange background such as these:





The symbols in current use are red diamond shapes. One department of the United Nations Organisation has developed these GHS labels for international use. 'GHS' is an abbreviation for Globally Harmonised System of Classification and Labelling of Chemicals, and the use of these labels is spreading throughout the world. **Table A** shows some of the more common ones, and includes a short description of their meanings.

SYMBOL	MEANING	
	Health hazard	includes warning on skin rashes, eye damage and ingestion
	Corrosive	can cause skin burns and permanent eye damage
	Flammable	can catch fire if heated or comes into contact with a flame
	Acute toxicity	can cause life-threatening effects, even in small quantities

table A Some pictograms used for chemical hazards.

In some cases, the substance may have more than one symbol, especially when it is an aqueous solution. For example, you are likely to use hydrochloric acid in three different concentrations:

- in a titration, it may have a concentration of about 0.1 mol dm^{-3}
- as a general laboratory reagent, it may have a concentration of 1 or 2 mol dm^{-3}
- for some purposes, you may use concentrated hydrochloric acid with a concentration of more than 10 mol dm^{-3}.

These very different concentrations have different hazards.

RISK ASSESSMENTS AND CONTROL MEASURES

The person in charge of a laboratory (or any other place of work) is responsible for making **risk assessments**. First, they look at the hazards of all the chemical substances, being guided by the hazard symbols. Then they consider the ways in which these substances will be used (this is assessing the risk). Finally, they write some guidelines for those who use the laboratory (these are the control measures).

EXAM HINT

If an exam question asks you to suggest a precaution when tackling a practical assignment, it is NOT enough to say 'wear a lab coat and goggles'. This should be standard practice for any practical assignment. You should comment on a particular procedure in the practical, such as, 'HCl gas is produced and so this step should be carried out in a fume cupboard'.

Such guidelines will consider many different factors, including:

- the amount used
- the age and experience of the person using it
- whether it will be heated
- whether ventilation or a fume cupboard should be used.

The control measures may refer to:

- the type of eye protection that should be worn
- the need to wear gloves
- keeping the cap on the bottle after removing some of the substance
- keeping the substance away from a source of heat
- what to do if some of the substance is spilled on the floor or gets on the skin.

Remember that there are also hazards and risks in your home. Many cleaning materials contain hazardous materials. **Fig A** shows a warning label for a household oven cleaner.

OVEN CLEANER
DANGER



HAZARD STATEMENTS:
Causes severe skin burns and eye damage

PRECAUTIONARY STATEMENTS:
- Wear protective gloves, protective clothing, eye protection, face protection.
- Wash hands thoroughly after handling.
- IF SWALLOWED: Rinse mouth Do NOT induce vomiting.
- IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses if present and easy to do; continue rinsing.
- IF ON SKIN: Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower.
- Dispose of contents/container in accordance with local regulations.

SEE SDS FOR MORE INFORMATION

▲ **fig** A You may find examples of hazards, risks and control measures in your home.

APPARATUS

Don't forget about the apparatus you might use in chemistry laboratories. Some of this apparatus may also have hazards.

For example, mercury thermometers obviously contain mercury, which is a hazardous substance. This is why you are more likely to use spirit thermometers and digital thermometers, their hazards are much lower.

Another example is methods of heating. Traditionally, heating involved the use of a Bunsen burner, tripod and gauze. Perhaps an electrical heating mantle would be safer, as well as easier to use.

In organic chemistry, some experiments involve glass apparatus for techniques such as distillation. A long time ago, the traditional way to connect the different pieces of glass apparatus was by using connecting glass tubing with corks or rubber bungs. More recently, apparatus made only of glass, connected mainly by ground glass joints, has been used. This is less hazardous because it does not involve the risks involved in assembling the apparatus.

LEARNING TIP

Consider the hazards of the substances you will use in an experiment of your choice. Decide what the risks of doing the experiment are, and suggest what control measures you think should be in place.

CHECKPOINT

1. Explain the difference between hazard and risk, using dilute sulfuric acid as an example.
2. A substance to be heated in an experiment is in a bottle labelled 'flammable'.

   Suggest two control measures that could be used to reduce the risk to a laboratory worker using this substance.

SUBJECT VOCABULARY

hazard something that could cause harm to a user

risk the chance of a hazard causing harm

risk assessment the identification of the hazards involved in carrying out a procedure and the control measures needed to reduce the risks from those hazards